

Proof. Let $x \geq -1$ be a real number and let n be a non-negative integer.

Base case $n = 0$. By definition,

$$(1+x)^0 = 1 \geq 1 + 0 \cdot x = 1.$$

Base case $n = 1$. Directly,

$$(1+x)^1 = 1+x \geq 1 + 1 \cdot x,$$

with equality for every admissible x .

Inductive hypothesis. Assume that for some integer $k \geq 1$ the inequality

$$(1+x)^k \geq 1+kx \tag{IH}$$

holds for all $x \geq -1$.

Inductive step. Multiply both sides of (IH) by the non-negative factor $1+x$ (note that $1+x \geq 0$ because $x \geq -1$). Using monotonicity of multiplication by a non-negative number we obtain

$$\begin{aligned} (1+x)^{k+1} &= (1+x)^k(1+x) \\ &\geq (1+kx)(1+x). \end{aligned} \tag{1}$$

Expanding the right-hand side gives

$$(1+kx)(1+x) = 1 + (k+1)x + kx^2. \tag{2}$$

Since $k \geq 1$ and $x^2 \geq 0$ for every real x , the term kx^2 is non-negative, i.e. $kx^2 \geq 0$. Hence

$$1 + (k+1)x + kx^2 \geq 1 + (k+1)x. \tag{3}$$

Combining (1), (2) and (3) yields

$$(1+x)^{k+1} \geq 1 + (k+1)x.$$

Thus the inequality holds for $k+1$.

Conclusion by induction. The statement is true for $n = 0$ and $n = 1$, and the inductive step shows that truth for $n = k$ implies truth for $n = k+1$. By the principle of mathematical induction, for every integer $n \geq 0$ and every real $x \geq -1$ we have

$$(1+x)^n \geq 1 + nx.$$

Equality condition.

- For $n = 0$ we have $(1+x)^0 = 1 = 1 + 0 \cdot x$ for all x .
- For $n = 1$ the inequality is an identity: $(1+x)^1 = 1+x$ for all x .
- For $n \geq 2$ equality in the inductive step can occur only when the non-negative term kx^2 vanishes, i.e. when $x = 0$. Consequently, for $n \geq 2$

$$(1+x)^n = 1 + nx \iff x = 0.$$

Hence, apart from the trivial cases $n = 0$ or $n = 1$, equality holds exactly when $x = 0$. $\square$